

IPCC AR6 WGI Chapter 7

The Earth's Energy Budget, Climate Feedbacks and Climate Sensitivity

Key Highlights for AI System Training — Table 7.15 GWP Values

Key Highlights to Bookmark:

- Table 7.15: GWP-100 values for CH₄ (29.8) and N₂O (273.0)
- Atmospheric lifetimes: CH₄ = 11.8 ± 1.8 years, N₂O = 109 ± 10 years
- Uncertainty ranges for GWP values
- AR6 updates vs AR5 values (CH₄ was 28 in AR5, now 29.8 in AR6)

Section 1: Table 7.15 — GWP-100 Values (Key Species)

Template Variable: {rag_context_gwp_values}

Table 7.15 provides emission metrics for selected species. GWP-100 (100-year Global Warming Potential) is the standard metric used in sustainability reporting under GRI 305, IFRS S2, and OJK frameworks. AR6 values supersede AR5 (2013) values.

Species	Formula	Lifetime (years)	GWP-100 (AR6)	GWP-100 (AR5)
Carbon Dioxide	CO ₂	variable*	1 (reference)	1
Methane	CH ₄	11.8 ± 1.8	29.8 ± 11	28
Nitrous Oxide	N ₂ O	109 ± 10	273 ± 118	265
HFC-134a	CF ₃ CH ₂ F	14.0	1,530	1,300
Sulphur Hexafluoride	SF ₆	3,200	25,200	23,500
Nitrogen Trifluoride	NF ₃	569	17,400	16,100

* CO₂ has no single lifetime; it is removed by multiple processes on different timescales (ocean uptake, land sinks, etc.)

Section 2: Spec Constants (Section 3.2)

These GWP values are hardcoded as constants in the ETL pipeline. The AI system uses them to convert activity data from CH₄ and N₂O into CO₂e. They MUST NOT be changed without formal review against the latest IPCC assessment.

Constant Name	Value	Source / Notes
GWP_CO2	1.0	Reference gas (by definition)
GWP_CH4	29.8	IPCC AR6 Table 7.15, 100-year, fossil origin (includes climate-carbon feedback)
GWP_N2O	273.0	IPCC AR6 Table 7.15, 100-year
GWP_SF6	25200	IPCC AR6 Table 7.15, 100-year (used for electrical equipment emissions)

CO₂e Conversion Formula

Total tCO₂e = tCO₂ + (tCH₄ x 29.8) + (tN₂O x 273.0) + (tSF₆ x 25200)

Key Differences: AR6 vs AR5

Gas	AR5 (2013)	AR6 (2021)	Impact
CH ₄	28	29.8 (+6.4%)	Slightly higher methane contribution to total CO ₂ e
N ₂ O	265	273 (+3.0%)	Slightly higher N ₂ O contribution
SF ₆	23,500	25,200 (+7.2%)	Higher SF ₆ impact for electrical equipment

VALIDATION CHECK

✓ Spec Section 3.2 GWP constants CONFIRMED matching IPCC AR6 Table 7.15: GWP_CH4 = 29.8, GWP_N2O = 273.0. These values are from the 100-year GWP with climate-carbon feedback (fossil CH₄). Source: IPCC, 2021, AR6, WGI, Chapter 7, Table 7.15. VALIDATED — spec and source document are in alignment.